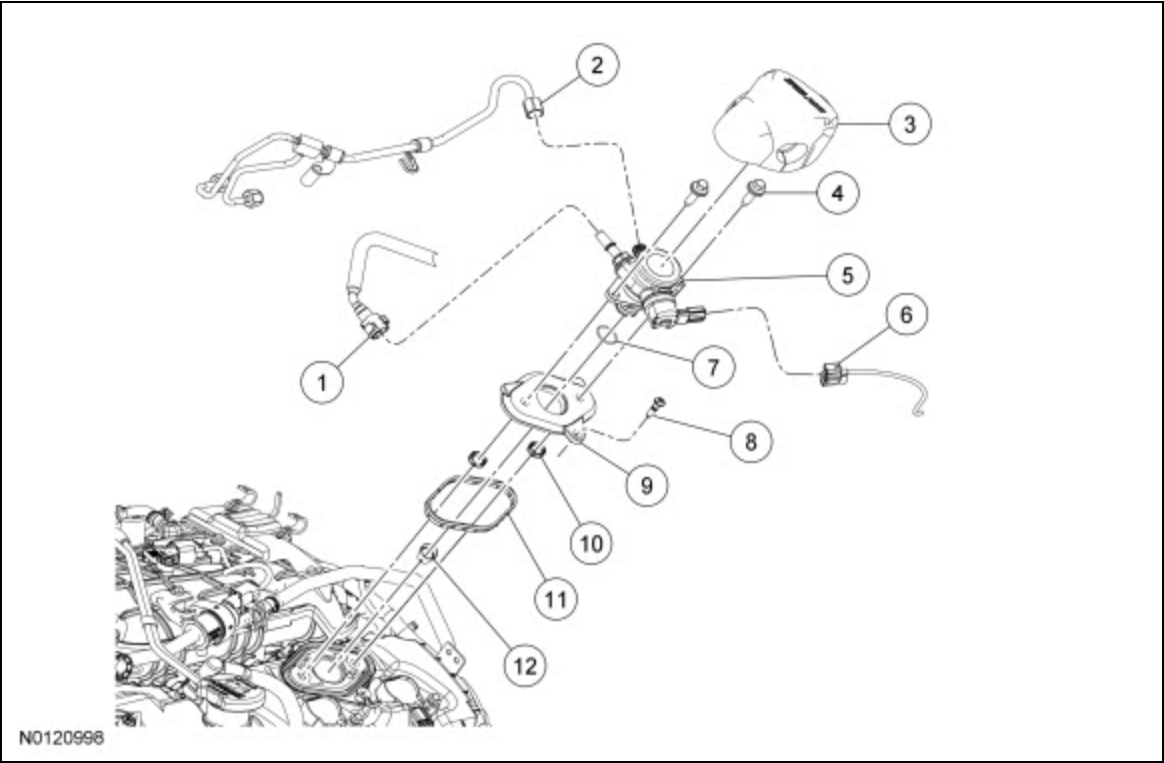


Fuel Injection Pump

 [Ford Ecat Electronic Parts Catalog](#)

Material

Item	Specification
Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS-M2C946-A



Item	Part Number	Description
1	—	Fuel jumper tube-to-fuel injection pump quick connect coupling (part of 9J280)
2	—	High pressure fuel tube-to-fuel injection pump flare nut (part of 9J323)
3	9K5409K540	Fuel injection pump noise insulator shield
4	W503297W503297	Fuel injection pump bolt (2 required)
5	93509350	Fuel injection pump
6	—	Fuel injection pump electrical connector (part of 12C581)
7	9E5839E583	Fuel injection pump O-ring seal
8	0081200812	Fuel injection pump mounting plate bolt
9	91789178	Fuel injection pump mounting plate

10	9374 9374	Fuel injection pump mounting plate O-ring seal (2 required)
11	9417 9417	Fuel injection pump mounting plate seal
12	9C587 9C587	Fuel injection pump tappet

Removal

WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

WARNING: Clean all fuel residue from the engine compartment. If not removed, fuel residue may ignite when the engine is returned to operation. Failure to follow this instruction may result in serious personal injury.

WARNING: Always disconnect the battery ground cable at the battery when working on an evaporative emission (EVAP) system or fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

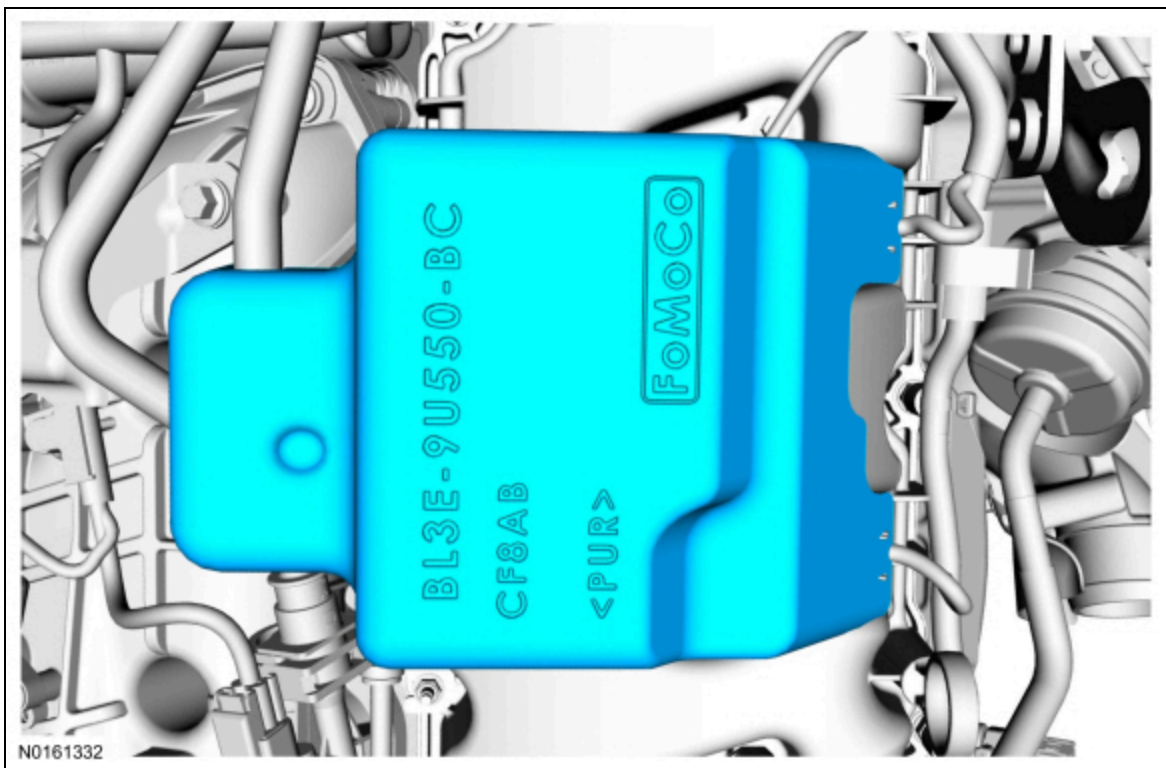
NOTICE: Do not loosen any fittings or plugs on the fuel injection pump.

NOTE: [Refer to the on-line Workshop Manual to view this illustration as an interactive exploded view, requires Adobe® Acrobat® 8.0 or higher.](#)

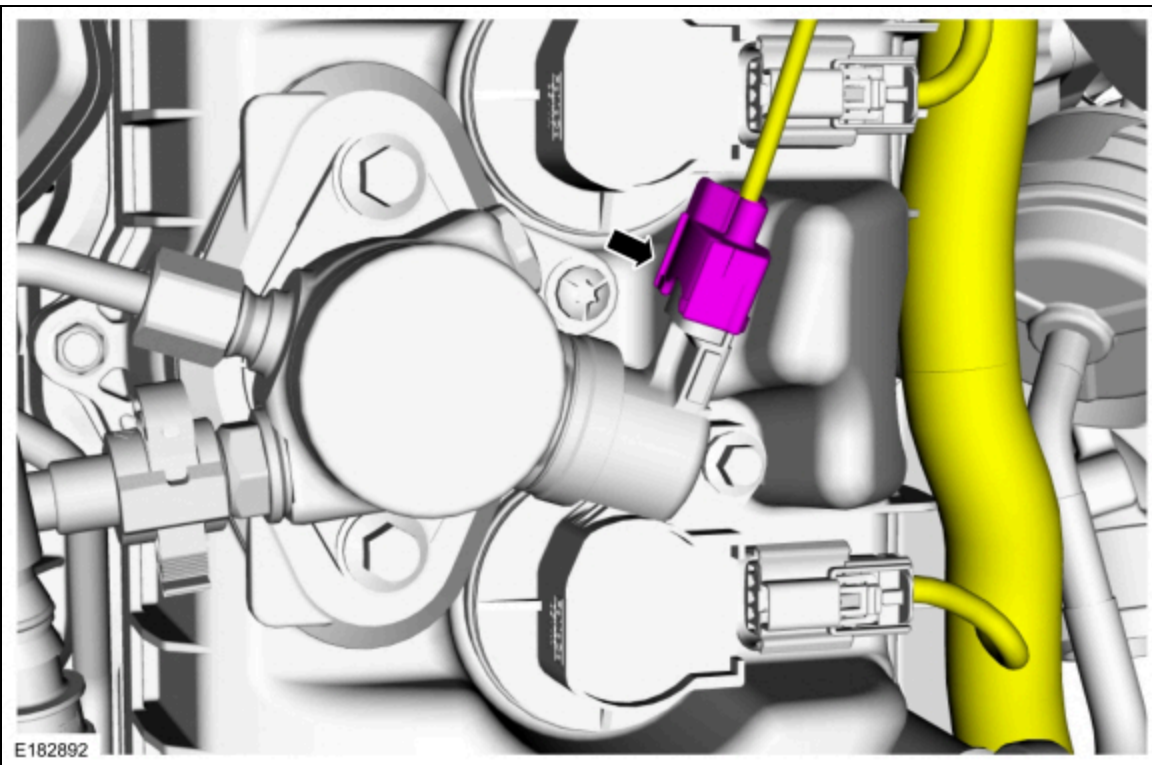
NOTE: [Refer to the on-line Workshop Manual to learn about using an Interactive Illustration.](#)

1. Release the fuel system pressure, refer to Section 310-00.
2. **NOTE:** Use care when removing the fuel injection pump noise insulator. Spreading the openings will aid in removing the fuel injection pump noise insulator and reduce the risk of damage.

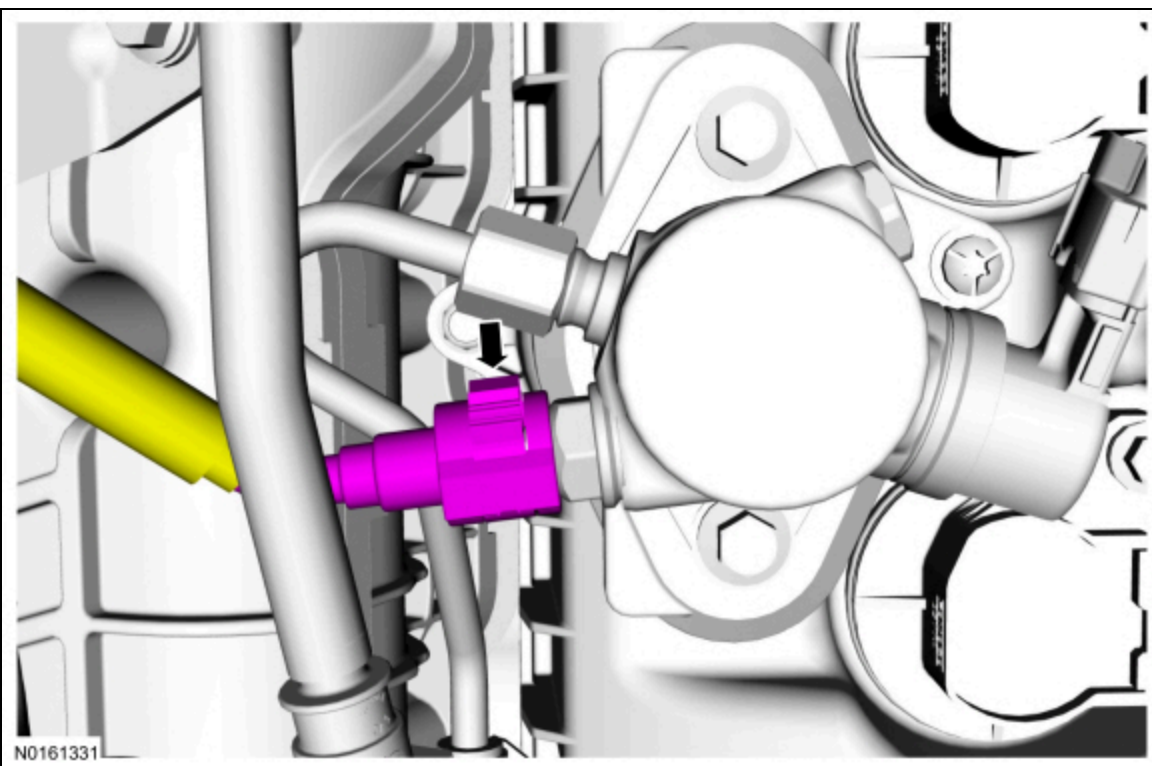
Remove the fuel injection pump noise insulator shield.



3. Disconnect the fuel injection pump electrical connector.



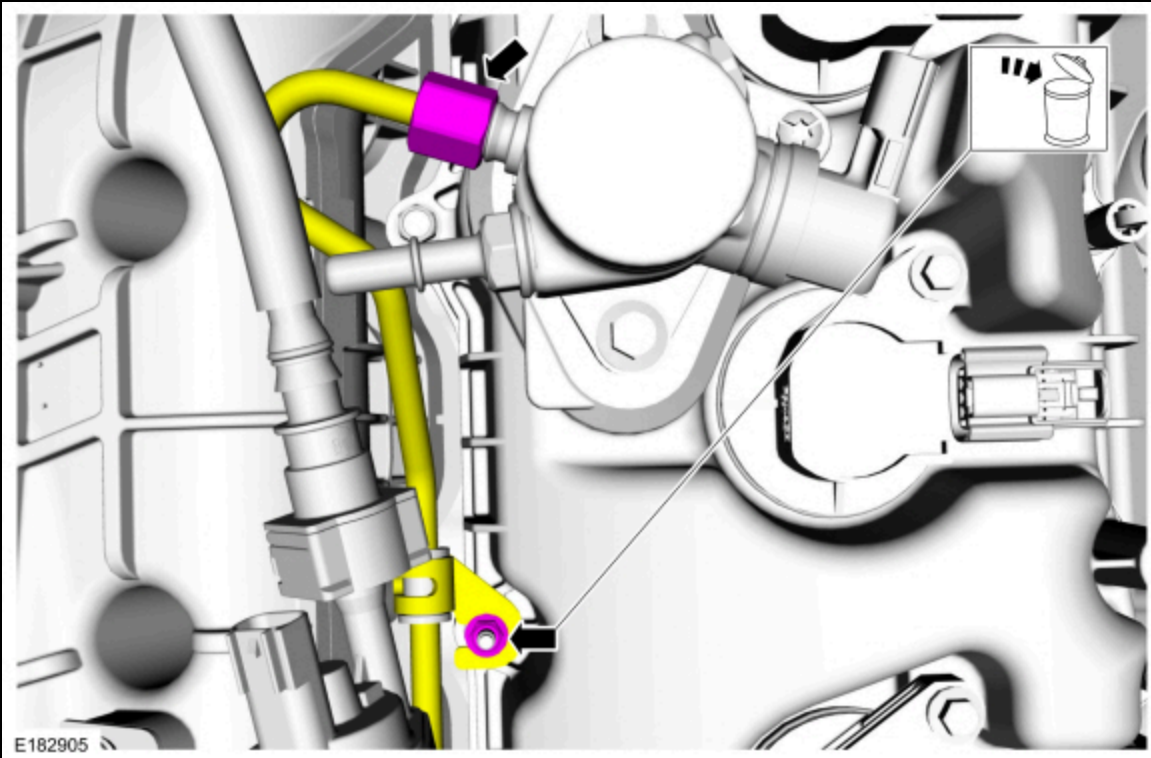
4. Disconnect the fuel jumper tube-to-fuel injection pump quick connect coupling, refer to Section 310-00.



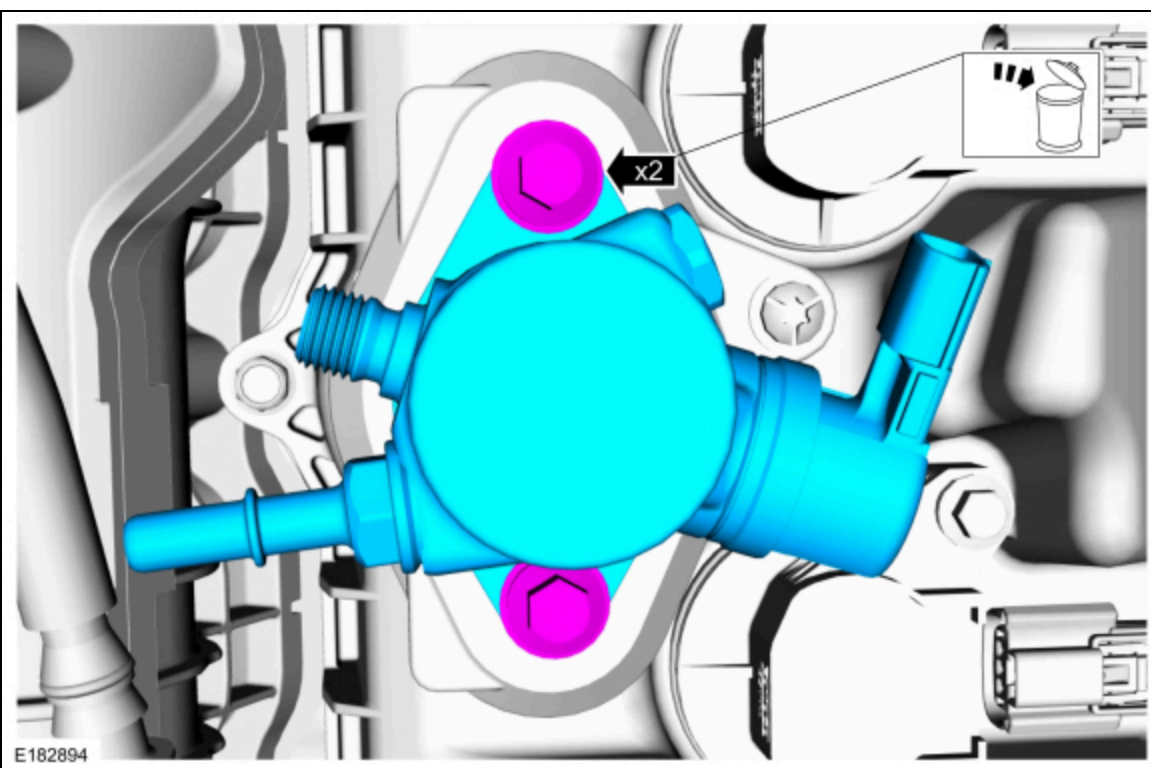
5. **NOTE:** To release the fuel pressure in the high pressure fuel tube, wrap the flare nut with a shop towel to absorb any residual fuel pressure during the loosening of the flare nut.

NOTE: When only disconnecting the high pressure fuel tube at the fuel injection pump flare nut, the high pressure fuel tube can be reused.

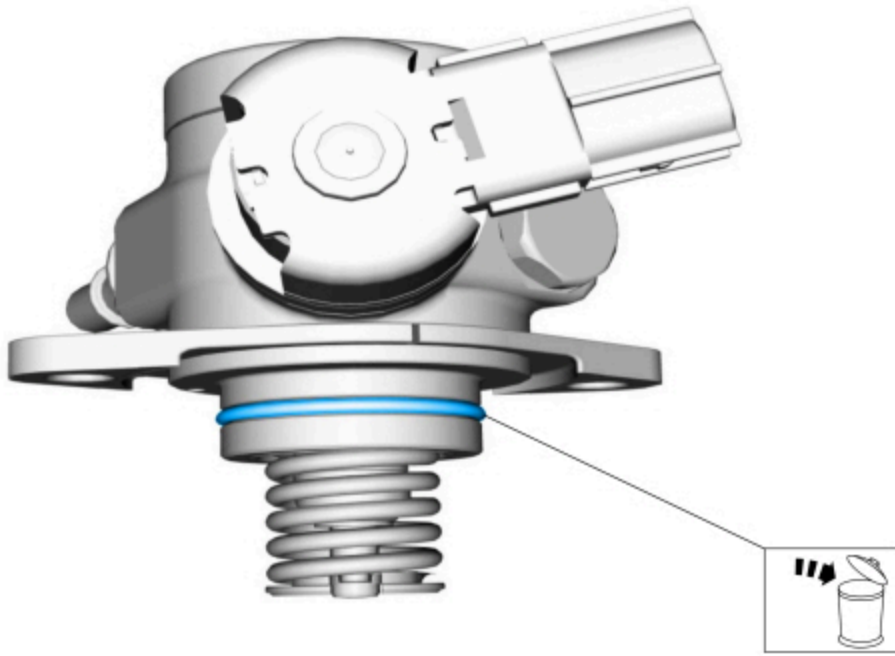
Remove the high pressure fuel tube bracket nut and discard. Disconnect the high pressure fuel tube-to-fuel injection pump flare nut.



6. Alternately loosen the 2 fuel injection pump bolts one complete revolution at a time.
- Discard the 2 fuel injection pump bolts.
 - Remove the fuel injection pump.



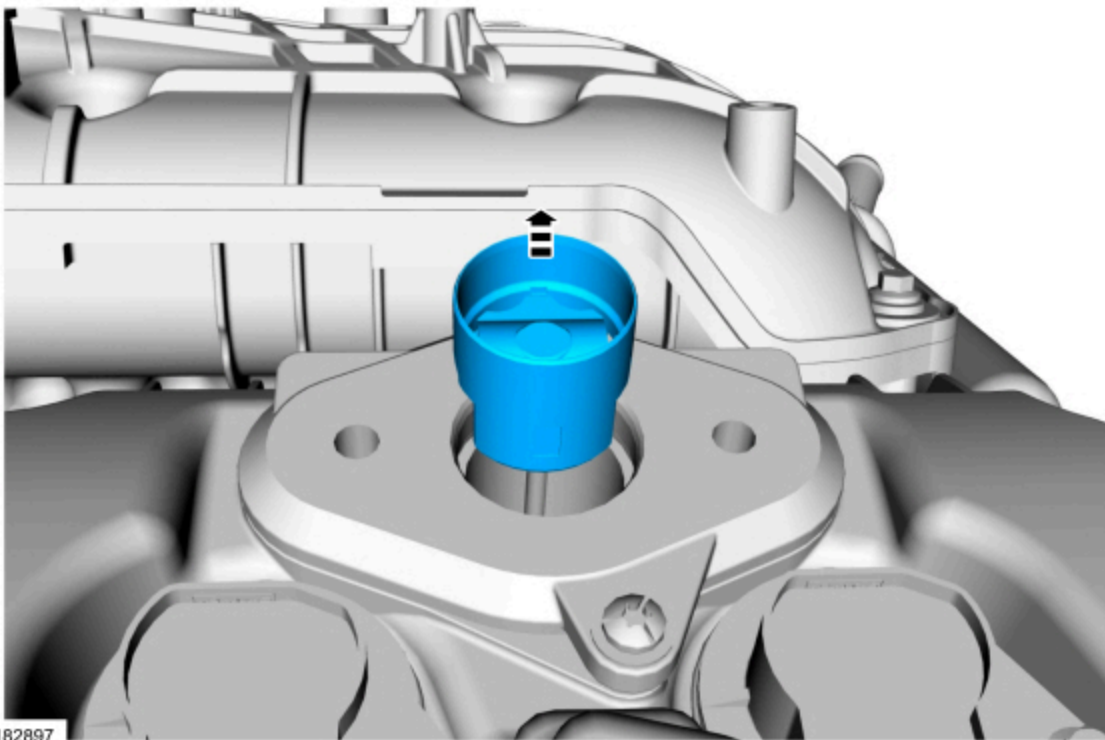
7. Remove the fuel injection pump O-ring seal and discard.



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8. Remove the fuel injection pump roller tappet.

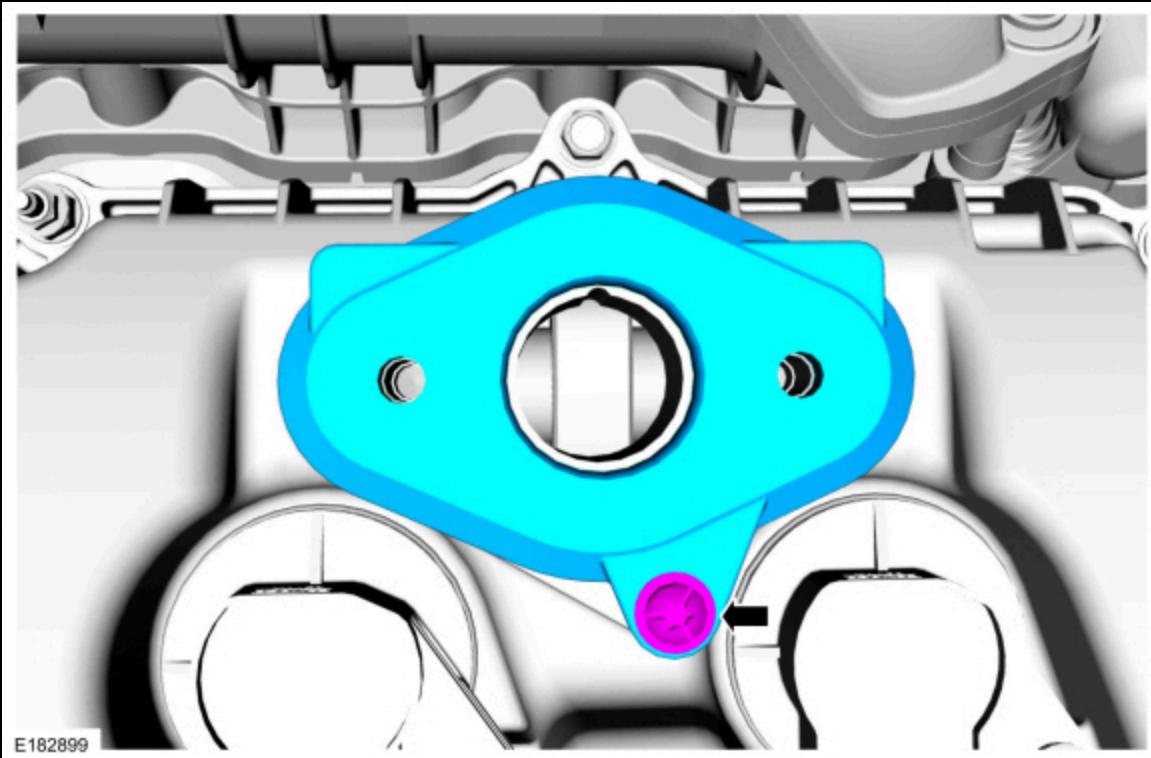
- Inspect the fuel injection pump roller tappet, refer to Fuel Injection Pump Roller Tappet Inspection in this section.



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Installation

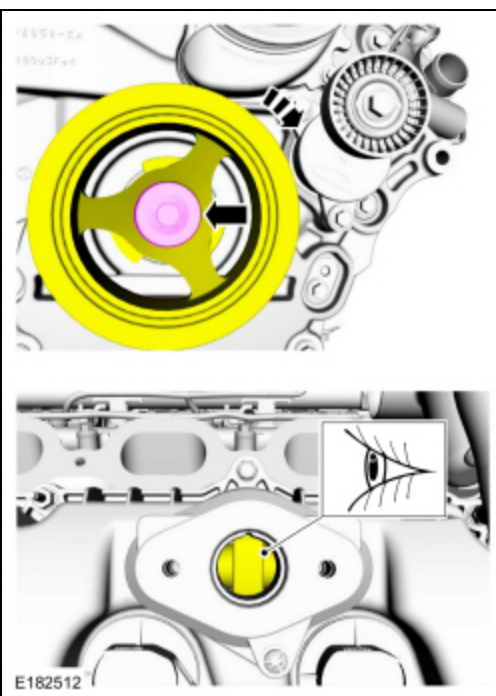
1. Remove the fuel injection pump mounting plate bolt and the fuel injection pump mounting plate.



2. NOTICE: Only rotate the crankshaft **CW** or damage to the engine may occur.

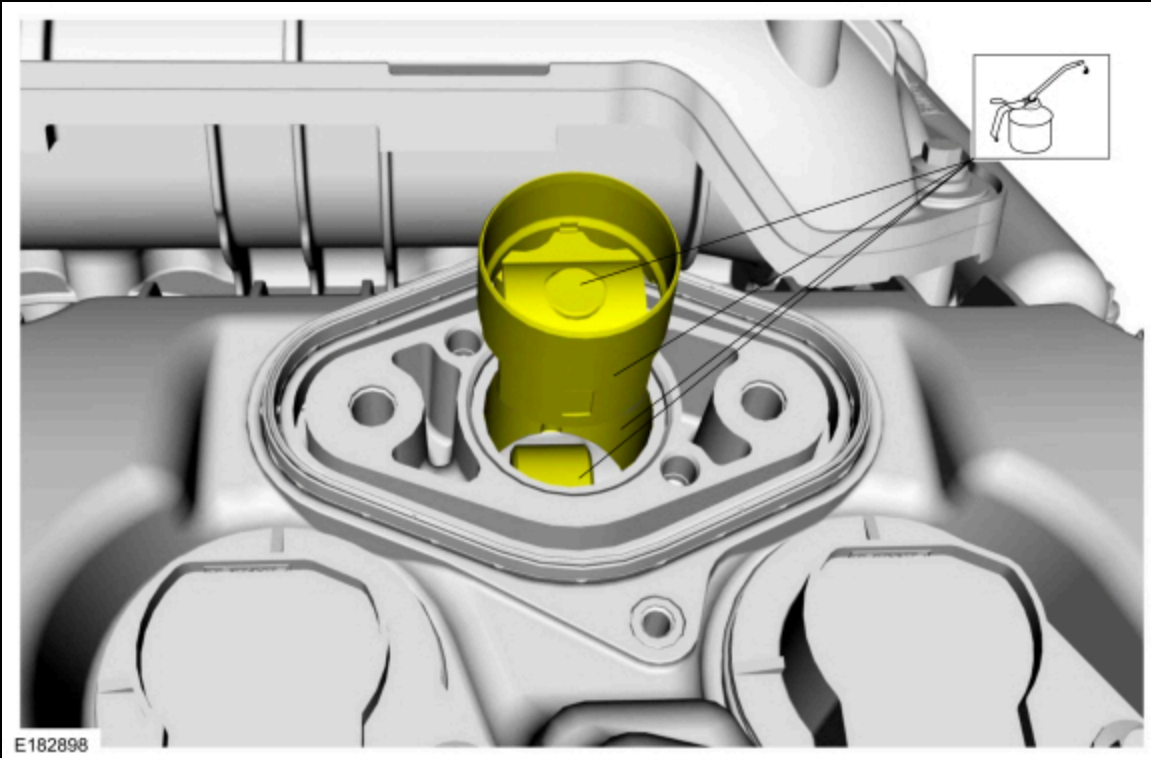
NOTICE: The cam lobe for the fuel injection pump must be at **Bottom Dead Center (BDC)** for the fuel injection pump installation.

Using the crankshaft pulley bolt, turn the crankshaft until the fuel injection pump cam lobe is at **BDC**.



3. NOTE: Material: Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada); or equivalent.

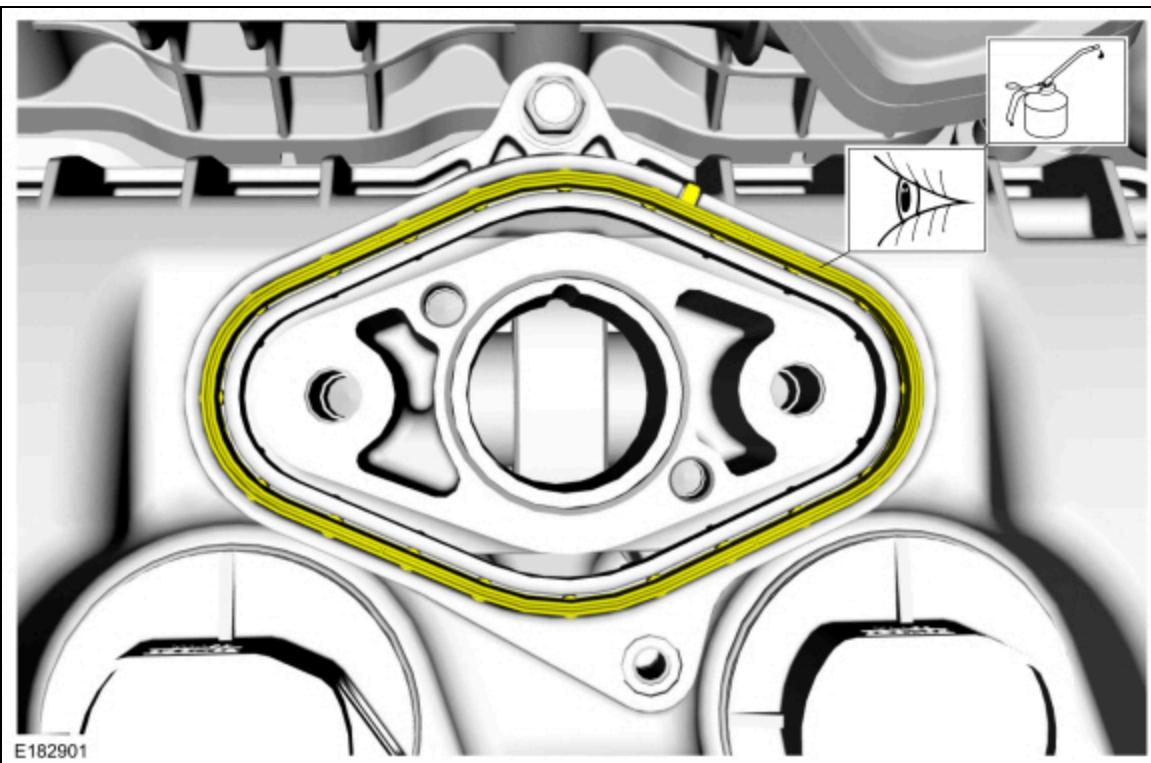
Apply clean engine oil to the fuel injection pump mounting pedestal bore, the camshaft lobe and the fuel injection pump roller tappet. Install the fuel injection pump roller tappet.



4. **NOTE:** Apply clean engine oil to the fuel injection pump mounting plate seal.

NOTE: Material: Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada); or equivalent.

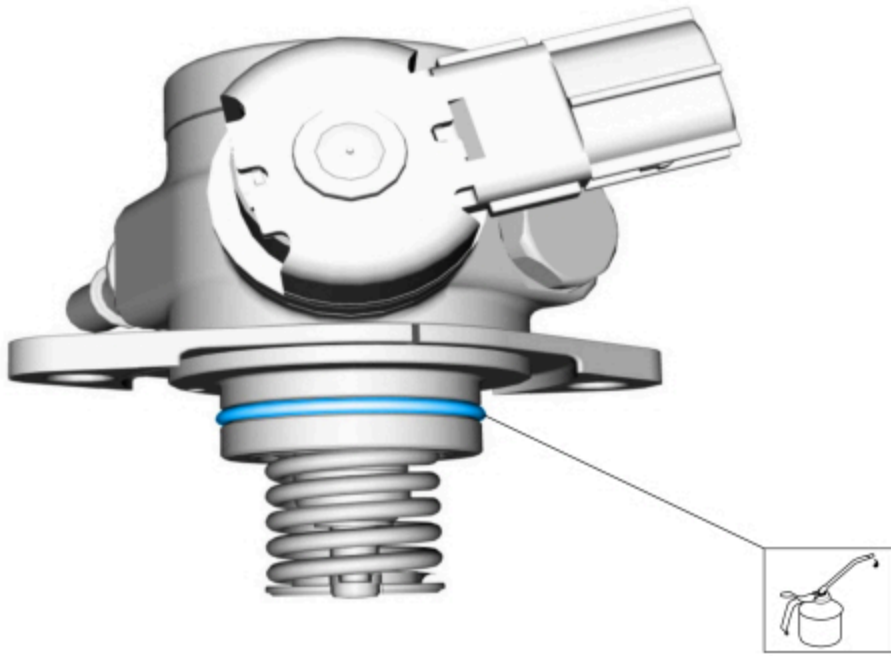
Inspect the fuel injection pump mounting plate seal and replace as necessary. Lubricate the fuel injection pump mounting plate seal with clean engine oil.



5. **NOTE:** Apply clean engine oil to the new fuel injection pump O-ring seal.

NOTE: Material: Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada); or equivalent.

Lubricate and install a new fuel injection pump O-ring seal onto the fuel injection pump. Lubricate the fuel injection pump mounting plate seal with clean engine oil.

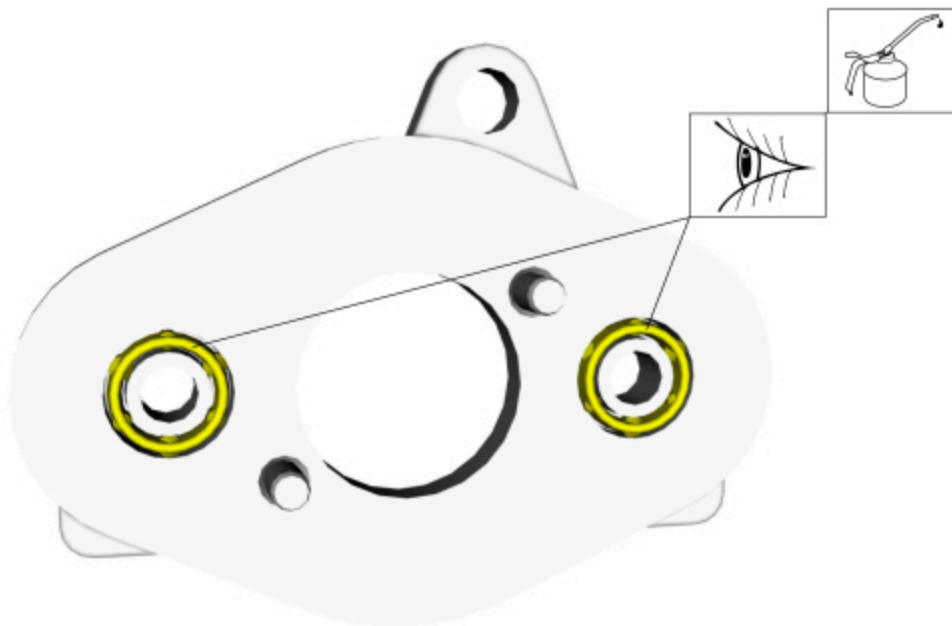


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6. **NOTE:** Apply clean engine oil to the fuel injection pump mounting plate O-ring seals.

NOTE: Material: Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada); or equivalent.

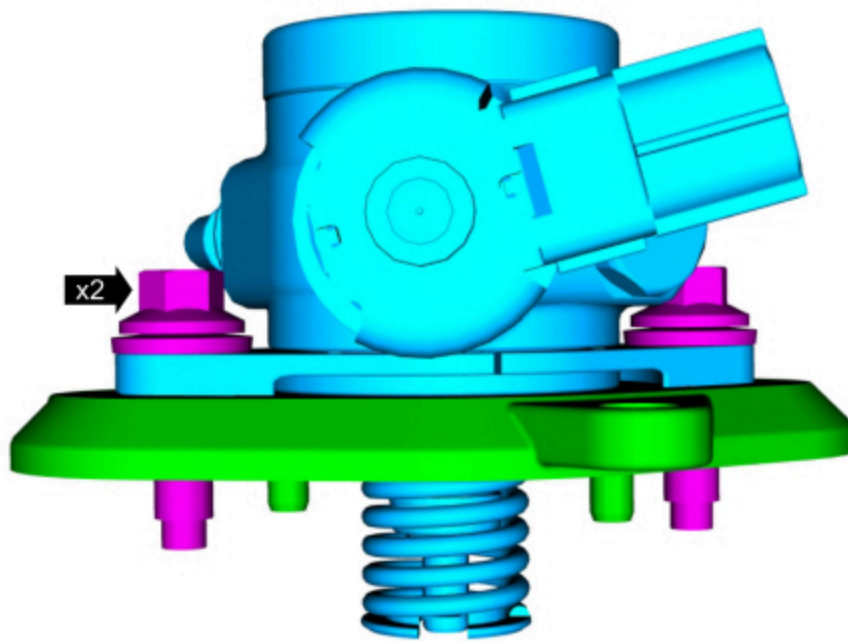
Inspect the fuel injection pump mounting plate O-ring seals and replace as necessary. Lubricate the fuel injection pump mounting plate O-ring seals with clean engine oil.



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7. **NOTE:** Make sure the mating surfaces are free of any dirt or foreign material.

Install the fuel injection pump mounting plate on to the fuel injection pump. Install the fuel injection pump bolts.



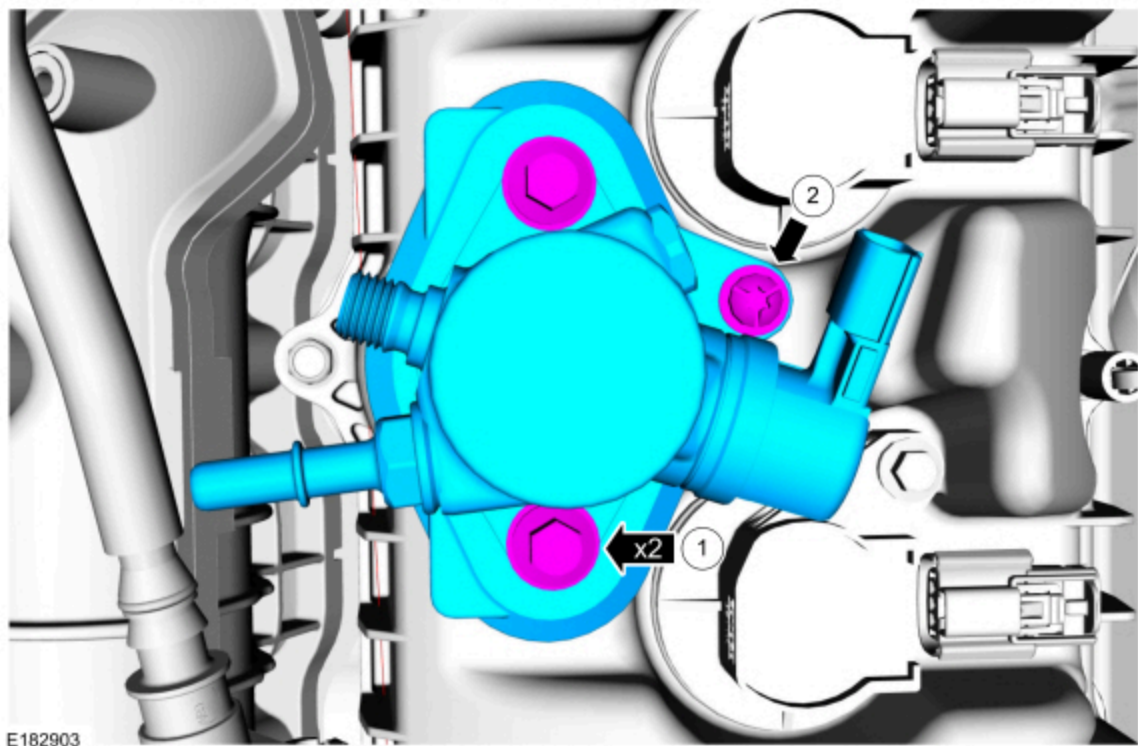
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8. Install the fuel injection pump with the fuel injection pump mounting plate, loosely install the new fuel injection pump bolts and loosely install the fuel injection pump plate bolt. Alternately tighten each fuel injection pump bolt one complete revolution until seated.

1. Tighten the 2 fuel injection pump bolts in the following 2 stages.

- Stage 1: tighten each bolt to 20 Nm (177 lb-in).
- Stage 2: tighten each bolt an additional 45 degrees.

2. Tighten the fuel injection pump mounting plate bolt to 10 Nm (89 lb-in).



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9. **NOTE:** Apply clean engine oil to the threads of the high-pressure fuel tube flare nut.

NOTE: Install the high pressure fuel tube flare nut and the new high pressure fuel tube bracket nut finger tight at this stage.

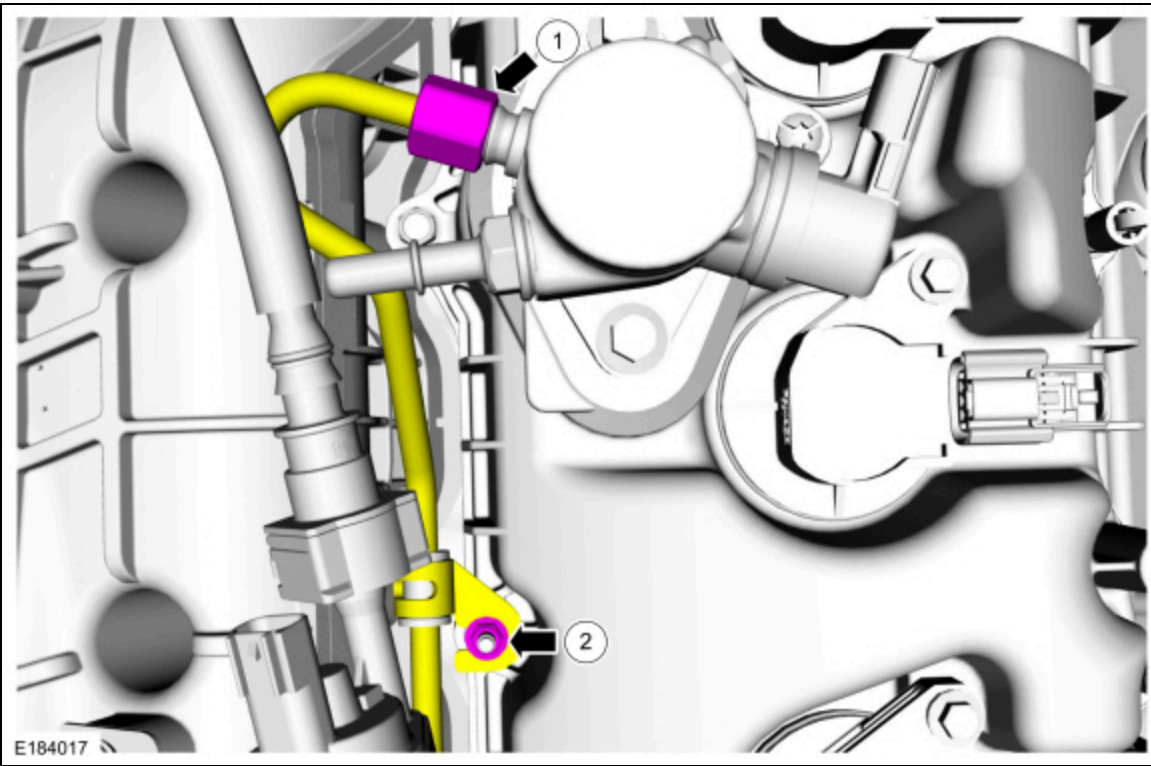
NOTE: Make sure that a new high pressure fuel tube bracket nut is installed.

NOTE: Calculate the correct torque wrench setting for the following torque using the Torque Wrench Adapter Formulas.

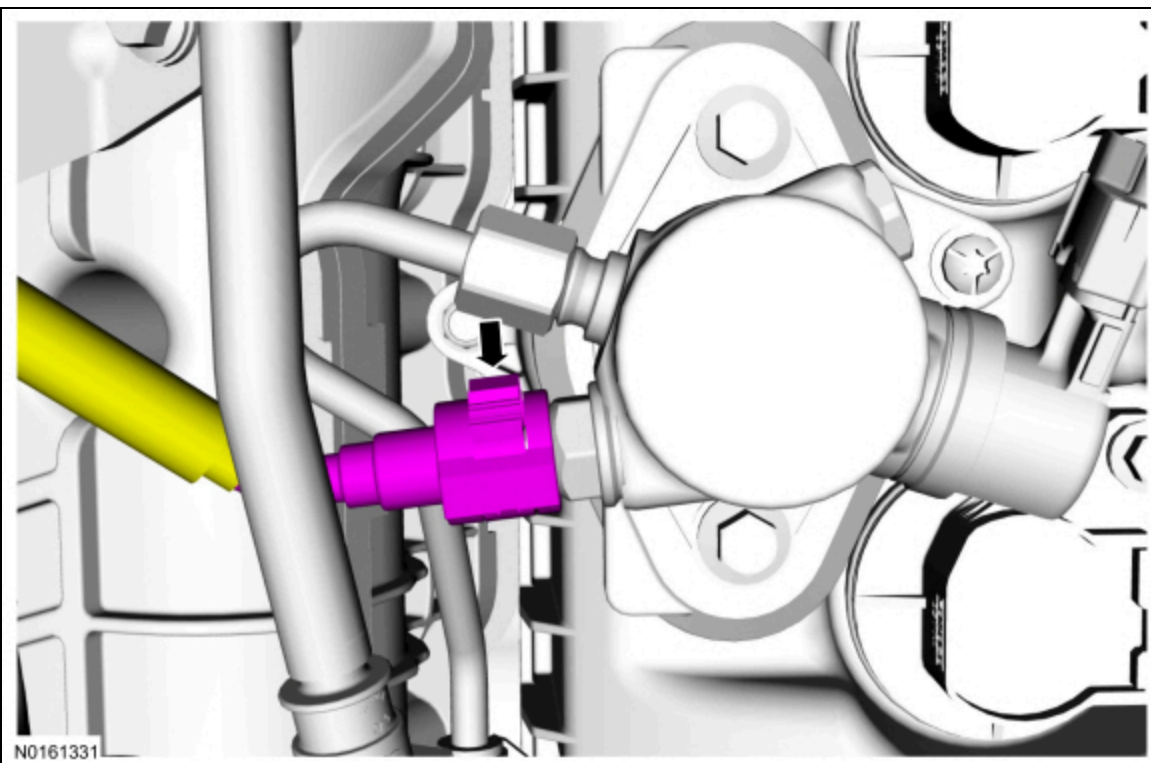
NOTE: Material: Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada); or equivalent.

Connect the high pressure fuel tube-to-fuel injection pump flare nut.

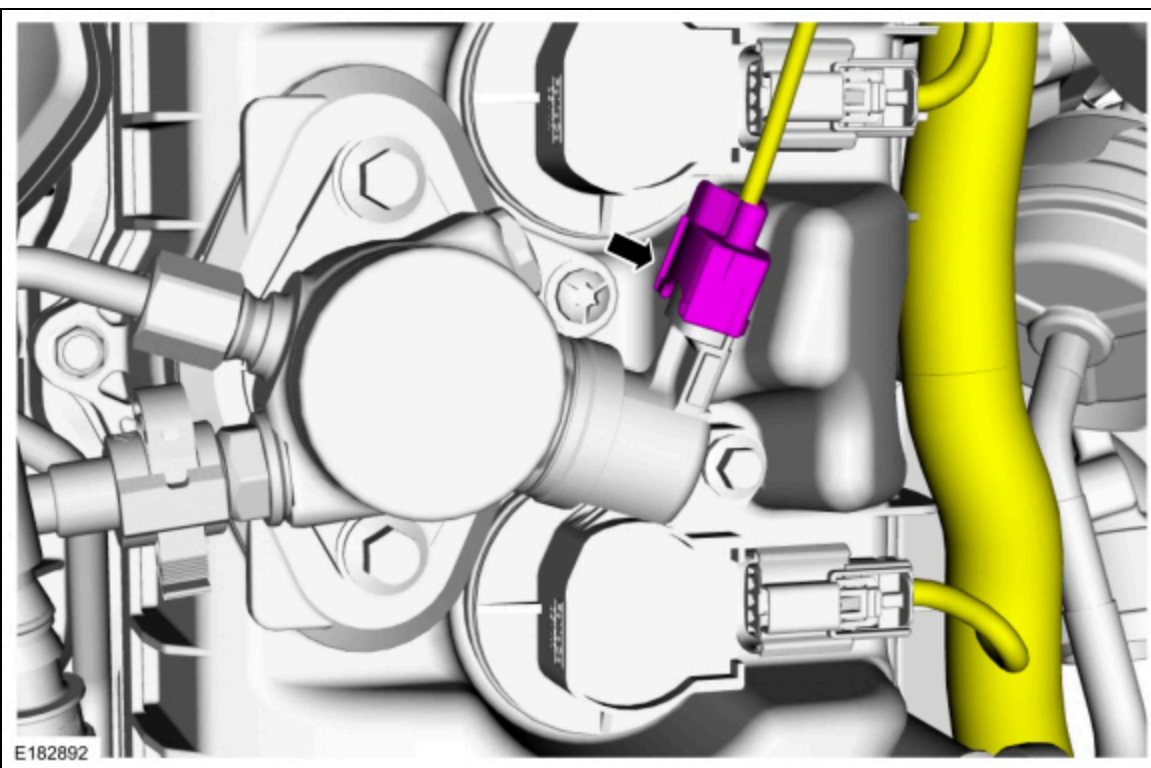
1. Tighten the high pressure fuel tube-to-high-pressure fuel pump flare nut.
 - Stage 1: Tighten to 15 Nm (133 lb-in).
 - Stage 2: Tighten an additional 30 degrees or 50 Nm (37 lb-ft), whichever comes first.
2. Tighten the new high pressure fuel tube bracket nut to 8 Nm (71 lb-in).



10. Connect the fuel jumper tube-to-fuel injection pump quick connect coupling. For additional information, refer to Section 310-00.



11. Connect the fuel injection pump electrical connector.



12. **NOTE:** Use care when installing the fuel injection pump noise insulator. Spreading the openings will aid in installing the fuel injection pump noise insulator and reduce the risk of damage.

Install the fuel injection pump noise insulator shield.

